FPPF HYPERCOVERINGS: QCOH IS WEAK SERRE AND A^*: D_QCOH(O_X) -> D_QCOH(O_U) IS AN EQUIVALENCE WITH QUASI-INVERSE RA_*

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: conditions (2) and (3) are delegated to two dedicated lemmas of the same chapter (lemma-descent-qcoh-for-fppf-hypercovering and lemma-cohomological-descent-qcoh-for-fppf-hypercovering) whose statements are verbatim the required criterion conditions - the author's modular structure, not an omission; the quasi-coherent characterization used is delivered as card L46152 of this batch.

Editorial note. The label lemma-coh-descent-qcoh-unbounded-for-fppf-hypercovering names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- P-coverings are flat, finitely presented and surjective. Hypercoverings chapter, Remark \label{hypercovering-remark-P-covering}. Lines 7679-7682: the maps a_n : U_n -> X and d_i^n : U_n -> U_{n-1} are flat, locally of finite presentation, surjective.
- Quasi-coherent = cartesian with quasi-coherent components on the total site. Same chapter, \label{lemma-quasi-coherent-sheaf} (card L46152 of this batch). Lines 7684-7687: the characterization of QCoh(O_U) inside Mod(O_U).
- Cartesian modules form a weak Serre subcategory. Same chapter, \label{lemma-Serre-subcat-cartesian-modules}. Lines 7688-7691: with flatness of the d_i^n shown above.
- QCoh on an algebraic space is a weak Serre subcategory. Properties of Spaces chapter, \label{spaces-properties-lemma-properties-quasi-coherent}. Lines 7692-7695: the component statement combined into QCoh(O_U) weak Serre.
- Cohomology on Sites, the unbounded equivalence criterion. Cohomology on Sites chapter, \label{sites-cohomology-lemma-equivalence-unbounded-one}. Lines 7700-7704: conditions (1)-(5) checked one by one.
- Flatness of the augmentation. Same chapter, \label{lemma-flat-augmentation-modules}. Line 7708: condition (1) from flatness of the a_n.

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- Descent of quasi-coherent modules for fppf hypercoverings. Same chapter, \label{lemma-descent-qcoh-for-fppf-hypercovering}. Lines 7711-7713: condition (2).
- Cohomological descent for quasi-coherent modules. Same chapter, \label{lemma-cohomological-descent-qcoh-for-fppf-hypercovering}. Lines 7716-7718: condition (3).
- Cohomology on the total site computes componentwise. Same chapter, \label{lemma-sanity-check-modules}. Lines 7745-7747: H^p(V/U_n, F) = H^p(V, F_n) for the vanishing in condition (4).
- Quasi-coherent cohomology vanishes on affines. Cohomology of Schemes chapter, \label{coherent-lemma-quasi-coherent-affine-cohomology-zero}; Cohomology of Spaces, Section \ref{spaces-cohomology-section-higher-direct-image}. Lines 7749-7754: H^p(V, F_n) = 0 for p > 0 with V affine.

Source reference. Original lemma 01XB: sources/stacks-chow/tags.txt, source lines 2512–2512. Prerequisite pointer invoked at proof line(s) 7758; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original remark 0DER: sources/stacks-chow/tags.txt, source lines 16811–16811. Prerequisite pointer invoked at proof line(s) 7693; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D7J: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 2669–2669. Prerequisite pointer invoked at proof line(s) 7699; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DHE: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 7628–7628. Prerequisite pointer invoked at proof line(s) 7726; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DHD: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 7544–7544. Prerequisite pointer invoked at proof line(s) 7722; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DH3: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 2286–2286. Prerequisite pointer invoked at proof line(s) 7718; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D7M: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 2817–2817. Prerequisite pointer invoked at proof line(s) 7698; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0DH2: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 2208–2208. Prerequisite pointer invoked at proof line(s) 7751; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0DE7: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 6985–6985. Prerequisite pointer invoked at proof line(s) 7687; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0D7V: sources/stacks-chow/tags.txt, source lines 16570–16570. Prerequisite pointer invoked at proof line(s) 7714; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0D6R: sources/stacks-chow/tags.txt, source lines 16531–16531. Prerequisite pointer invoked at proof line(s) 7733; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 071Y: sources/stacks-chow/tags.txt, source lines 9007–9007. Prerequisite pointer invoked at proof line(s) 7756; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 03M1: sources/stacks-chow/tags.txt, source lines 4733–4733. Prerequisite pointer invoked at proof line(s) 7706; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let X be an algebraic space over S. Let U be a simplicial algebraic space over S. Let $a : U \to X$ be an augmentation. Assume $a : U \to X$ is an fppf hypercovering of X. Then $QCoh(\mathcal{O}_U)$ is a weak Serre subcategory of $Mod(\mathcal{O}_U)$ and*

$$a^* : D_{QCoh}(\mathcal{O}_X) \longrightarrow D_{QCoh}(\mathcal{O}_U)$$

is an equivalence of categories with quasi-inverse given by Ra_*. Here $a : Sh(U_{étale}) \to Sh(X_{étale})$ is as in Section 0DE7.

Proof. First observe that the maps $a_n : U_n \to X$ and $d_i^n : U_n \to U_{n-1}$ are flat, locally of finite presentation, and surjective by Hypercoverings, Remark 0DER.

Recall that an $\mathcal{O}_U$-module $\mathcal{F}$ is quasi-coherent if and only if it is cartesian and $\mathcal{F}_n$ is quasi-coherent for all n. See Lemma 0D7M. By Lemma 0D7J (and flatness of the maps $d_i^n : U_n \to U_{n-1}$ shown above) the cartesian modules for a weak Serre subcategory of $Mod(\mathcal{O}_U)$. On the other hand $QCoh(\mathcal{O}_{U_n}) \subset Mod(\mathcal{O}_{U_n})$ is a weak Serre subcategory for each n (Properties of Spaces, Lemma 03M1). Combined we see that $QCoh(\mathcal{O}_U) \subset Mod(\mathcal{O}_U)$ is a weak Serre subcategory.

To finish the proof we check the conditions (1) – (5) of Cohomology on Sites, Lemma 0D7V one by one.

Ad (1). This holds since a_n flat (seen above) implies a is flat by Lemma 0DH3.

Ad (2). This is the content of Lemma 0DHD.

Ad (3). This is the content of Lemma 0DHE.

Ad (4). Recall that we can use either the site $U_{étale}$ or $U_{spaces,étale}$ to define the small étale topos $Sh(U_{étale})$, see Section 0DE7. The assumption of Cohomology on Sites, Situation 0D6R holds for the triple $(U_{spaces,étale}, \mathcal{O}_U, QCoh(\mathcal{O}_U))$ and by the same reasoning for the triple $(U_{étale}, \mathcal{O}_U, QCoh(\mathcal{O}_U))$. Namely, take

$$\mathcal{B} \subset \mathrm{Ob}(U_{étale}) \subset \mathrm{Ob}(U_{spaces,étale})$$

to be the set of affine objects. For $V/U_n \in \mathcal{B}$ take $d_{V/U_n} = 0$ and take Cov_{V/U_n} to be the set of étale coverings $\{V_i \to V\}$ with V_i affine. Then we get the desired vanishing because for $\mathcal{F} \in QCoh(\mathcal{O}_U)$ and any $V/U_n \in \mathcal{B}$ we have

$$H^p(V/U_n, \mathcal{F}) = H^p(V, \mathcal{F}_n)$$

by Lemma 0DH2. Here on the right hand side we have the cohomology of the quasi-coherent sheaf $\mathcal{F}_n$ on U_n over the affine object V of $U_{n,étale}$. This vanishes for $p > 0$ by the discussion in Cohomology of Spaces, Section 071Y and Cohomology of Schemes, Lemma 01XB.

Ad (5). Follows by taking $\mathcal{B} \subset \mathrm{Ob}(X_{spaces,étale})$ the set of affine objects and the references given above. □